

DBMS – PRODUCT AND CATEGORY MANAGEMENT

Project Title: Zara-Fashion E-Commerce Product and Category Management System

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint
Category_ID	NUMBER	Primary Key
Category_Name	VARCHAR2(100)	NOT NULL, UNIQUE
Description	VARCHAR2(200)	—

1.2 Product Table

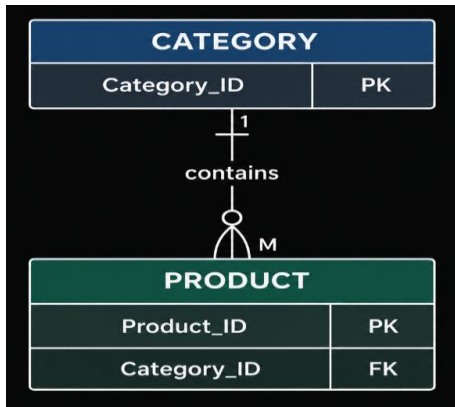
Attribute	Data Type	Constraint
Product_ID	NUMBER	Primary Key
Product_Name	VARCHAR2(100)	NOT NULL
Brand	VARCHAR2(50)	NOT NULL
Price	NUMBER(10,2)	NOT NULL
Size	VARCHAR2(20)	—
Colour	VARCHAR2(30)	—
Category_ID	NUMBER	Foreign Key

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key

- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key
- Foreign Key references Category.Category_ID
- Relationship: One-to-Many (1:M)



3. Table creation

3.1 Create category table

```

SQL> CREATE TABLE Category (
2     Category_ID NUMBER PRIMARY KEY,
3     Category_Name VARCHAR2(100) NOT NULL UNIQUE,
4     Description VARCHAR2(200)
5 );

Table created.
  
```

3.2 Create Product Table

```

SQL> CREATE TABLE Product (
2     Product_ID NUMBER PRIMARY KEY,
3     Product_Name VARCHAR2(100) NOT NULL,
4     Brand VARCHAR2(50) NOT NULL,
5     Product_Size VARCHAR2(20),
6     Colour VARCHAR2(30),
7     Price NUMBER(10,2) NOT NULL,
8     Category_ID NUMBER,
9     CONSTRAINT fk_product_category
10    FOREIGN KEY (Category_ID)
11    REFERENCES Category(Category_ID)
12 );

Table created.
  
```

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO Category
  2 VALUES (1, 'Dresses', 'Fashionable dresses for women');

1 row created.

SQL>
SQL> INSERT INTO Category
  2 VALUES (2, 'T-Shirts', 'Casual and comfortable t-shirts');

1 row created.

SQL>
SQL> INSERT INTO Category
  2 VALUES (3, 'Jeans', 'Stylish denim jeans');

1 row created.

SQL>
SQL> INSERT INTO Category
  2 VALUES (4, 'Shoes', 'Fashion and casual footwear');

1 row created.

SQL>
SQL> INSERT INTO Category
  2 VALUES (5, 'Accessories', 'Fashion accessories');
```

4.2 Product Records

```
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Product_Size, Colour, Price, Category_ID)
  3 VALUES
  4 (101, 'Floral Dress', 'Zara', 'M', 'Black', 2499.00, 1);

1 row created.

SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Product_Size, Colour, Price, Category_ID)
  3 VALUES
  4 (102, 'Casual T-Shirt', 'Zara', 'L', 'White', 999.00, 2);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Product_Size, Colour, Price, Category_ID)
  3 VALUES
  4 (103, 'Slim Fit Jeans', 'Zara', 'M', 'Blue', 1999.00, 3);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Product_Size, Colour, Price, Category_ID)
  3 VALUES
  4 (104, 'Running Shoes', 'Zara', '8', 'White', 2999.00, 4);

1 row created.

SQL>
SQL> INSERT INTO Product
  2 (Product_ID, Product_Name, Brand, Product_Size, Colour, Price, Category_ID)
  3 VALUES
  4 (105, 'Leather Belt', 'Zara', 'M', 'Brown', 1299.00, 5);

1 row created.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM Category;
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
1
```

```
Dresses
```

```
Fashionable dresses for women
```

```
2
```

```
T-Shirts
```

```
Casual and comfortable t-shirts
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
3
```

```
Jeans
```

```
Stylish denim jeans
```

```
4
```

```
Shoes
```

```
CATEGORY_ID
```

```
CATEGORY_NAME
```

```
DESCRIPTION
```

```
Fashion and casual footwear
```

```
5
```

5.2 Display Product Records

```
SQL> SELECT * FROM Product;
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
BRAND
```

```
PRODUCT_SIZE
```

```
COLOUR
```

```
PRICE
```

```
CATEGORY_ID
```

```
101
```

```
Floral Dress
```

```
Zara
```

```
Black
```

```
2499
```

```
M
```

```
1
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
BRAND
```

```
PRODUCT_SIZE
```

```
COLOUR
```

```
PRICE
```

```
CATEGORY_ID
```

```
102
```

```
Casual T-Shirt
```

```
Zara
```

```
White
```

```
999
```

```
L
```

```
2
```

```
PRODUCT_ID
```

```
PRODUCT_NAME
```

```
BRAND
```

```
PRODUCT_SIZE
```

```
COLOUR
```

```
PRICE
```

```
CATEGORY_ID
```

```
103
```

```
Slim Fit Jeans
```

```
Zara
```

```
M
```

6. Product Update

6.1.Update the price of Floral Dress.

```
SQL> UPDATE Product
  2 SET Price = 2699.00
  3 WHERE Product_ID = 101;
```

1 row updated.

```
SQL> SELECT * FROM Product
  2 WHERE Product_ID = 101;
```

PRODUCT_ID	PRODUCT_NAME	BRAND	PRODUCT_SIZE	COLOUR	PRICE	CATEGORY_ID
101	Floral Dress	Zara	M	Black	2699	1

6.2 Update Size and Colour

```
SQL> UPDATE Product
  2 SET Product_Size = 'L',
  3     Colour = 'Black'
  4 WHERE Product_ID = 103;
```

1 row updated.

7. Product Deletion

7.1.Delete Product – Hand Bag

```
SQL> DELETE FROM Product
  2 WHERE Product_ID = 110;
```

1 row deleted.

7.2 Delete Product – Running Shoes

```
SQL> DELETE FROM Product
  2 WHERE Product_ID = 104;
```

1 row deleted.

8. Category-wise Product Reports

8.1 Product Listing by Category

```

SQL> SELECT
  2     C.Category_Name,
  3     P.Product_Name,
  4     P.Brand,
  5     P.Price,
  6     P.Product_Size,
  7     P.Colour
  8 FROM Category C
  9 JOIN Product P
 10 ON C.Category_ID = P.Category_ID
 11 ORDER BY C.Category_Name;

```

CATEGORY_NAME

PRODUCT_NAME

BRAND

PRICE

PRODUCT_SIZE

COLOUR

Accessories

Leather Belt

Zara

1299

M

Brown

CATEGORY_NAME

PRODUCT_NAME

BRAND

PRICE

PRODUCT_SIZE

COLOUR

Dresses

Summer Dress

Zara

2799

S

Pink

CATEGORY_NAME

8.2 Products in Dresses Category

```

SQL> SELECT
  2     P.Product_ID,
  3     P.Product_Name,
  4     P.Brand,
  5     P.Price,
  6     P.Product_Size,
  7     P.Colour
  8 FROM Product P
  9 JOIN Category C
 10 ON P.Category_ID = C.Category_ID
 11 WHERE C.Category_Name = 'Dresses';

```

PRODUCT_ID

PRODUCT_NAME

BRAND

PRICE

PRODUCT_SIZE

COLOUR

101

Floral Dress

Zara

2699

M

Black

PRODUCT_ID

PRODUCT_NAME

BRAND

PRICE

PRODUCT_SIZE

COLOUR

106

Summer Dress

Zara

2799

S

Pink

8.3 Product Count by Category

```
SQL> SELECT
  2     C.Category_Name,
  3     COUNT(P.Product_ID) AS Product_Count
  4 FROM Category C
  5 LEFT JOIN Product P
  6 ON C.Category_ID = P.Category_ID
  7 GROUP BY C.Category_Name
  8 ORDER BY C.Category_Name;
```

CATEGORY_NAME

PRODUCT_COUNT

Accessories
1

Dresses
2

Jeans
2

CATEGORY_NAME

PRODUCT_COUNT

Shoes
1

T-Shirts
2